Branch-and-Price Redistricting as an Audit-Ready Column-Generation Contract

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Draft — May 11, 2026

Abstract

This draft presents the BISECT branch-and-price path as a staged exact optimization implementation. It describes pricing, master-problem, bound, exact fixture, and RPLAN/RCTX audit-package contracts without claiming mature large-instance branch-price performance. The paper treats disconnected columns, partial pools, and formulation-only outputs as reportable statuses rather than silent solver failures.

1 Introduction

U.17 documents `bisect-column` and `bisect-cli::exact_cmd`, exposed as `bisect exact --method branch-and-price`. The paper is an exact-optimization implementation paper about column-generation contracts, not a claim that the large-instance pricing strategy is complete.

2 Implementation Contract

The branch-and-price surface has three contracts: a pricing contract that generates connected balanced columns or explicit failure statuses, a master contract that reports objective and bound state, and a package contract that turns solved fixture outputs into final-plan artifacts.

The key reproducibility requirement is that a reader can tell which loop stage produced the artifact. A pricing artifact is not a solved partition; a master report is not necessarily a branch-and-price proof; and an exact fixture package is scoped to the fixture model, objective, and column universe recorded in the run context.

3 Audit Artifacts

Solved outputs should converge to the RPLAN/RCTX fixed point with certificate and manifest sidecars. Column-generation-specific evidence remains in the run context and solver report; the final RPLAN carries the assignment, not the entire proof history.

Contract	Must report	Evidence level
Pricing	column validity, balance, reduced-cost status	L0 fixture
Master	selected columns, objective, bound status	L0/L1 exact package
Package	final assignments, manifest, certificate	L1 audit path

Table 1: Branch-and-price staged implementation contracts.

Status	Meaning	Allowed claim
Pricing-valid	connected balanced columns generated	pricing contract works on fixture
Pricing-failed	no valid column or explicit fallback	failure is reproducible
Master-bounded	objective and bound reported	scoped bound under column pool
Fixture-solved	exact fixture package emitted	exact fixture result
Formulation-only	model/report built without solve	staged implementation only

Table 2: Status vocabulary for branch-and-price outputs.

The audit package should identify the column pool, selected columns, master objective, fixture or instance id, and certificate profile. This keeps the generic RPLAN assignment separate from the specialized column-generation evidence needed to interpret exactness.

The public golden package can be checked either through the RPLAN certificate verifier or through the BISECT manifest verifier.

4 Evaluation Plan

The near-term evidence is L0 pricing/master tests, an L1 exact package test, and a public U.17 golden package in the package corpus. Comparisons with branch-and-cut, formulation-only reports, and heuristic construction are future work until command transcripts and larger fixture outputs are attached.

5 Limitations

The current branch-and-price story is intentionally fixture-scale. Branching strategy, large-instance pricing, and broad runtime claims remain staged. The paper should not describe a formulation-only report or partial column pool as a solved exact instance.

The paper also does not claim that the generated column universe is complete unless the solver report and fixture contract say so. Pricing failures, duplicate columns, and disconnected candidates must remain visible because they are part of the implementation evidence.

Claim	Current evidence	Publication gap
Pricing emits valid columns	L0 pricing fixture tests	column-pool transcript
Master reports objective/bound	L0/L1 master package tests	solver version and output log
Solved outputs audit through RPLAN	L1 exact package test and public golden package	larger solver-produced instance
Large-instance branch-price scales	future evidence placeholder	benchmark protocol

Table 3: Evidence ladder for U.17 claims.

6 Conclusion

U.17 establishes the column-generation audit contract: pricing and master artifacts may be specialized, but selected final plans should still reach the same RPLAN/RCTX verification path as other U-series algorithms.

This lets the implementation mature in public stages. The draft can claim a fixture-scale contract now, while reserving large-instance exact performance for future solver transcripts and benchmark evidence.

References

BISECT project. U.17 branch-and-price redistricting paper plan, 2026. Local planning document and implementation roadmap.